

Preference Discovery and Consumer Search: Evidence from Checkout Abandonment

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Introduction

Consider yourself buying a new smartphone.

Consumer Search

- ▶ Large selection of products, features and prices.
- ▶ Need to invest time to find a good match.
- ▶ Trade off: buy the best product searched so far, or continue searching.

Preference Discovery

- ▶ At first, you may not fully know your price sensitivity to smartphones.
- ▶ You may only learn about this as you search.

Introduction

- ▶ Sellers often use tactics to get consumers to buy early:
 - ▶ Low stock warnings (“12 people are viewing this, only 2 products left in stock”).
 - ▶ Countdown clocks (“deal ends in 00:07:32”).
 - ▶ One-click purchase buttons.
- ▶ Not knowing your true price sensitivity can have large welfare consequences.

In the context of smartphones, we ask:

- ▶ How much learning takes place?
- ▶ How large are the welfare losses from not fully knowing one's preferences?

This Paper

What we do:

- ▶ Use detailed click-stream data from a popular website in Central Asia.
- ▶ Unique features of the store allow us to disentangle search and preference discovery.
- ▶ Observe that consumers become more price sensitive after abandoning a checkout.
- ▶ Estimate a structural model of search with learning about price sensitivity.

Main results:

- ▶ Consumers 10% more price sensitive at purchase compared to when starting search.
- ▶ Average own-price elasticities:
 - ▶ Under standard search model without learning: -1.4 .
 - ▶ In our model with learning: -2.2 .
- ▶ One click purchase button: $\uparrow$ seller revenue, $\downarrow$ consumer welfare.

Related Literature and Contributions

- ▶ Well-established literature on consumer search but without preference discovery.
 - ▶ e.g. Kim et al. (2010), Ursu (2018).
- ▶ Experimental studies with learning about preferences.
 - ▶ e.g. Plott and Zeiler (2005), Cao and Zhang (2021).
- ▶ Some empirical applications of preference discovery without search.
 - ▶ e.g. Narita (2018), Grenet et al. (2022) in school/university matching systems.
- ▶ Theoretical work combining search and learning about preferences.
 - ▶ Dzyabura and Hauser (2019)
- ▶ $\Rightarrow$ We are the first to model search and learning in an empirical setting.

Introduction: Empirical Setting

- ▶ Our data are from an e-commerce website in a Central Asian country.
 - ▶ Platform has a 70% market share.
 - ▶ Over 60% of the population has an account.
- ▶ Website is run by a Fintech company:
 - ▶ Consumers must registered and have a bank account with them to make a purchase.
- ▶ Each data point is a time-stamped consumer click on a product, which is either a view, a checkout or a purchase.
- ▶ We focus on the smartphone category, and collect characteristics of each product (price, brand, RAM, storage, etc.).
 - ▶ Raw data: 104m clicks, 7.6m users, 1,707 unique products.

Product list page:

Price

☐

10,000 - 49,999 ₺ (83)

☐

50,000 - 99,999 ₺ (225)

☐

100,000 - 149,999 ₺ (115)

☐

150,000 - 199,999 ₺ (58)

☐

200,000 - 499,999 ₺ (278)

☐

more than 500,000 ₺ (104)

Brand name

☐

ASUS

☐

Apple

☐

BQ

☐

Blackview

☐

Caterpillar

☐

Gionee

show more

Rating

★★★★★and up (788)

★★★★★and up (800)

★★★★★and up (800)

★★★★★and up (803)

Popular models

☐

Apple iPhone 11 (20)

☐

Apple iPhone 12 Pro Max (24)

☐

Apple iPhone Xr (24)

☐

Xiaomi Redmi (37)

☐

Xiaomi Redmi Note (45)

0-0-12

10% ₺



Apple iPhone 11 128Gb Slim Box Black

★★★★★(1375 reviews)

Price

By installments

334 790 ₺

27 900 ₺ x12

0-0-12

10% ₺



Xiaomi Redmi 9A 2 / 32Gb gray

★★★★★(1361 reviews)

Price

By installments

43 990 ₺

3 666 ₺ x12

0-0-12

10% ₺



Apple iPhone 12 Pro 128Gb Blue

★★★★★(762 reviews)

Price

By installments

515 500 ₺

42 959 ₺ x12

0-0-12

10% ₺



Xiaomi Redmi 9T 4 / 128Gb gray

★★★★★(745 reviews)

Price

By installments

78 000 ₺

6 500 ₺ x12

0-0-12

10% ₺



Samsung Galaxy A32 4 / 128Gb black

★★★★★(380 reviews)

Price

By installments

104 300 ₺

8 692 ₺ x12

0-0-12

10% ₺



Apple iPhone 12 Pro Max 256Gb Blue

★★★★★(505 reviews)

Price

By installments

597 000 ₺

49 750 ₺ x12

0-0-12

10% ₺



Apple iPhone 12 Pro Max 256Gb Blue

★★★★★(505 reviews)

Price

By installments

597 000 ₺

49 750 ₺ x12

0-0-12

10% ₺



Apple iPhone 12 Pro Max 256Gb Blue

★★★★★(505 reviews)

Price

By installments

597 000 ₺

49 750 ₺ x12

0-0-12

10% ₺



Apple iPhone 12 Pro Max 256Gb Blue

★★★★★(505 reviews)

Price

By installments

597 000 ₺

49 750 ₺ x12

Viewing a product shows all information on delivery cost/times, reviews, specifications:



0-0-12

10% ₪

Smartphone Apple
iPhone 11 128Gb Slim
Box Black

★★★★★ (1375 reviews)

Price | By installments

334 890 ₪ | 27 908 ₪ x 12 months

Select a seller

- nfc technology: yes
- screen type: color IPS, Liquid Retina HD
- diagonal: 6.1 inch
- RAM size: 4096 MB
- internal memory: 128 GB
- battery capacity: 3110 mAh

To enlarge, click on the picture



Sellers | Reviews | Specifications

By installments for

	3 months	6 months	12 months	24 months
Di_Mobile ★★★★☆ (45 reviews)				
 Pickup  Delivery : free ,tomorrow	334 580 ₪	27 882 ₪	Select	
FORA ★★★★★ (5165 reviews)				
 Pickup  Delivery : free ,tomorrow	334 890 ₪	27 908 ₪	Select	
Evrika ★★★★★ (3426 reviews)				
 Pickup  Delivery : free ,September 30th	334 890 ₪	27 908 ₪	Select	
Dream ★★★★★ (22695 reviews)				
 Pickup  Delivery : free ,tomorrow	334 890 ₪	27 908 ₪	Select	

Pressing “Select” brings you straight to the checkout (no shopping cart!):

① Delivery method

123, 123, 123

Delivery - free **tomorrow**

Change

② Payment method

☐ Spend Bonuses

☒  Kaspi Gold

☐  Kaspi Red

☐  By installments

3 months

6 months

12 months

24 months

Payment per month

27 882 ₸

Checkout



Apple iPhone 11 128Gb Slim Box Black

Salesman Di_Mobile

Quantity:

- 1 +

 PCS.

The price of the product 334 580 ₸
Delivery is free

To pay 334 580 ₸

- ▶ No new information is provided at the checkout.
- ▶ Just one more button to finalize the purchase.
- ▶ Yet, 59% of checkouts are abandoned in our data.

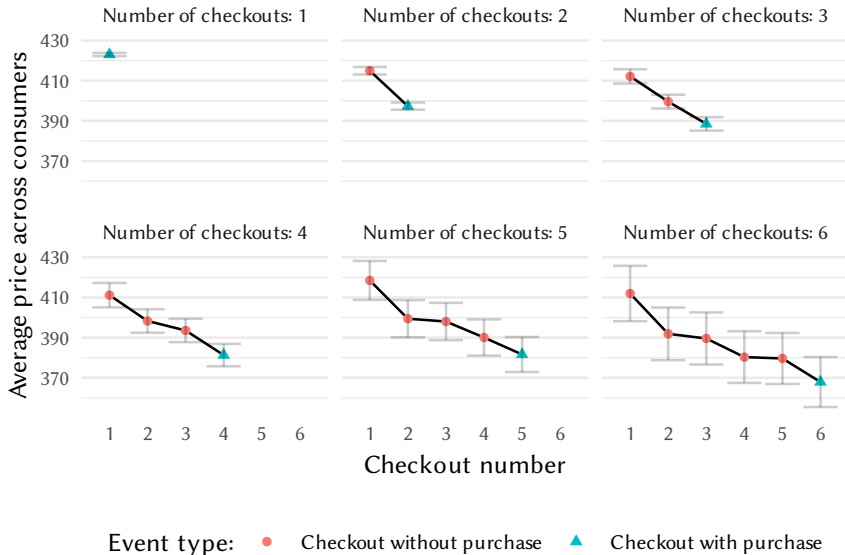
Checkout Abandonment

- ▶ Many common reasons for checkout abandonment don't apply here:
 - ▶ Learning shipping times, delivery fees, other fees.
 - ▶ Effort of account creation, inputting payment details.
 - ▶ Unfamiliarity or lack of trust with the site.

Our interpretation:

- ▶ When searching, consumers compare many characteristics *between* products.
- ▶ They then bring the product perceived to be the best to the checkout.
- ▶ Then they are faced with a simple decision: *buy this or not*.
 - ▶ This focuses their attention on price only.
 - ▶ They reconsider their price sensitivity.
 - ▶ If this changes, it can make the selected product no longer optimal $\Rightarrow$ abandon.

Descriptive Evidence of Consumers Becoming More Price Sensitive



Causal Impact of Abandonment on Next Checkout Price

- ▶ Example search paths for 2 consumers:

Step	Consumer 1 (“Treated”)	Consumer 2 (“Control”)
1	View <i>A</i>	View <i>A</i>
2	View <i>B</i>	View <i>B</i>
3	View <i>C</i>	View <i>C</i>
4	View, checkout & abandon <i>D</i>	View <i>D</i>
5	View <i>E</i>	View <i>F</i>
6	View & checkout <i>G</i>	View & checkout <i>H</i>

- ▶ *Idea*: Both consumers arguably similar ex ante, but consumer 1 abandons in step 4.
- ▶ *Goal*: estimate impact of abandonment on price of the next checked-out product.
- ▶ *Method*: Regress price on treatment and search-sequence×search-length FEs.
- ▶ *Result*: Treated units check out products on average \$13.04 cheaper (p -val 0.033).
 - ▶ Mean dependent variable: \$381.01.

Model Overview

1. Consumer starts with prior beliefs about their price sensitivity.
2. With their current beliefs, they search products to obtain their match value.
3. Once an optimal choice is made, the consumer brings the product to the checkout.
4. They receive a signal about their true price sensitivity and update their beliefs.
5. With the posterior beliefs:
 - ▶ If the product is still optimal with posterior beliefs: *finalize purchase*.
 - ▶ If it is no longer optimal with posterior beliefs: *abandon checkout; go back to step 2*.

Setup

- ▶ Consumer i chooses between products j from the set $\mathcal{M}_i$.
- ▶ Consumer i 's true utility from product j is:

$$u_{ij} = \underbrace{\beta_i}_{\text{price sensitivity}} \times \underbrace{p_{ij}}_{\text{price}} + \underbrace{\mathbf{x}_j' \boldsymbol{\gamma}}_{\text{value of product characteristics}} + \underbrace{\varepsilon_{ij}}_{\text{match value}}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

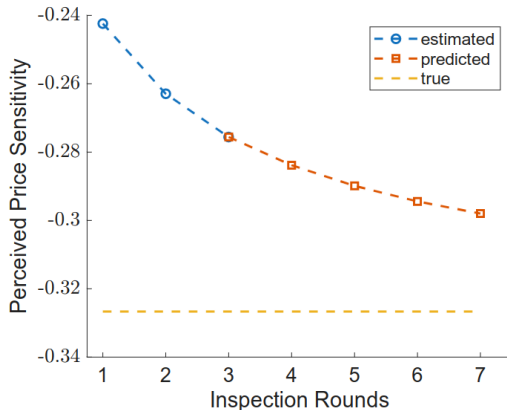
- ▶ Consumers know p_{ij} and $\mathbf{x}_j$, but pay a search cost to learn ε_{ij} .
- ▶ Let r denote the “round” of search, where each checkout starts a new round.
- ▶ In round r , the consumer has beliefs $\mathcal{N}(\beta_{ir}, \omega_r^2)$ on their price sensitivity.
- ▶ The initial prior mean is determined by:

$$\beta_{i1} \sim \mathcal{N}(\beta_i + \delta, \sigma_0^2)$$

- ▶ At the checkout, consumer receives a signal $\beta_i^s \sim \mathcal{N}(\beta_i, \sigma_s^2)$.
- ▶ With the signal, Bayesian updating to get β_{ir+1} and ω_{r+1}^2 .

Results

- ▶ Consumers initially underestimate their price sensitivity by 25.8%
- ▶ At purchase, it is 10% higher compared to beginning search on average.



- ▶ Standard Weitzman model (w/o learning) has price elasticities 36% smaller.

One-Click Purchase Counterfactual

- ▶ We consider a counterfactual where the website has a one-click purchase button.
- ▶ If they click, they do not receive a signal and always complete the purchase.
- ▶ Assume those who checkout click with button with 10% probability (exogenous).
- ▶ Find seller revenue $\uparrow$ 10% but consumer welfare decreases.

Conclusion

- ▶ We show evidence that consumers become more price sensitive following a checkout abandonment.
- ▶ We estimate a model of search and learning about one's price sensitivity.
- ▶ Our counterfactual shows that a one-click purchase button can lower consumer welfare.

Causal Impact of Abandonment on Next Checkout Price: Details

- ▶ *Approach:* Match consumers who:

1. view the same products in the same order in the first $k \geq 4$ clicks, and
2. checkout a product exactly ℓ clicks later.

where some consumers checkout but abandon the product in step k .

- ▶ Regress price of product checked out in step $k + \ell$ on:

- ▶ Indicator for abandoning checkout in step k .
- ▶ product-sequence $\times$ search-length fixed effects

- ▶ 3,204 product-sequence $\times$ search-length combinations with a match.

- ▶ 13,683 matched consumers (3661 treated, 10,022 control).

- ▶ *Result:* Checked-out product in step $k + \ell$ is on average \$13.04 cheaper (p -val 0.033).

- ▶ Mean dependent variable: \$381.01. [Back](#)